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MILITARY SPECIFICATION

HEATER, VEHICULAR COMPARTMENT: MULTIFUEL BURNING, 24 VOLT, D.C.

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers multifuel burning heaters, of 24-volt D.C. nominal rating, used with compression ignition and spark ignition engines for heating crew and engine compartments of military vehicles.

1.2 Classification. Heaters shall be of the following capacities, as specified (see 6.2):

Capacity A - 20,000 BTU/HR
Capacity B - 30,000 BTU/HR
Capacity C - 60,000 BTU/HR

1.2.1 Dimensions. Heaters shall conform to the dimensions shown on the applicable specification sheet. The arrangement of exhaust, inlet, and outlet connections shall be as shown on the specification sheet (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: DRSTA-GSS, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 2540

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-P-416	- Plating, Cadmium (Electrodeposited).
TT-C-490	- Cleaning Method and Pretreatment of Ferrous Surfaces for Organic Coatings.
TT-E-485	- Enamel, Semigloss, Rust-Inhibiting.
TT-E-529	- Enamel, Alkyd, Semigloss.
TT-P-636	- Primer Coating, Alkyd, Wood and Ferrous Metal.
VV-F-800	- Fuel Oil, Diesel.
VV-G-1690	- Gasoline, Automotive, Leaded or Unleaded.

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MIL-P-116	- Preservation, Methods of.
MIL-P-514	- Plate, Identification, Instruction and Marking, Blank.
MIL-G-3056	- Gasoline, Automotive, Combat.
MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-P-14105	- Paint, Heat-resisting (for Steel Surfaces).

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

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MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity, or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Qualification. The heaters furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4. and 6.).

3.2 Materials. Materials shall be as specified herein and in referenced specifications and standards. Material shall be free from defects which adversely affect performance or serviceability of the finished product (see 6.5).

3.2.1 Plating and painting. Unless otherwise specified, coating and painting shall be as follows: All interior steel parts, other than stainless steel, shall be cadmium plated in accordance with type I, class 2 of QQ-P-416. The heat exchanger housing shall be painted with high temperature olive drab paint conforming to MIL-P-14105. All other external metal painted parts shall receive initial coat in accordance with TT-P-636 or TT-E-485, and finish coat of olive drab color in accordance with TT-E-529 or TT-E-485.

3.3 Design and construction.

3.3.1 Construction. Heaters shall fit into current vehicle applications without modification to heater or vehicular components. Heaters shall be completely interchangeable as a unit with heaters of other manufacture used in the same application.

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3.3.1.1 Interchangeability. All parts having the same manufacturer or ordnance part number shall be completely interchangeable with respect to installation and performance.

3.3.2 Electrical. When tested as specified in 4.8.5, the current shall be not more than the values specified in table I.

TABLE I. Current.

Capacity (see 1.2)	-65°F Amperes	
	Start	Run
A	20.0	8.0
B	20.0	11.0
C	23.0	16.0

3.3.2.1 Voltage. The heaters shall start and operate satisfactorily at an applied voltage of 20 to 28.5 Volts direct current (d.c.).

3.3.2.2 Controls. When specified (see 6.2), heaters shall be furnished complete with controls to start, stop, and operate heater.

3.3.3 Fuel. The heaters shall be capable of performing as specified herein using fuel in accordance with VV-F-800, VV-G-1690, MIL-G-3056, or MIL-T-5624 as applicable. When testing as specified in 3.4, the fuel used shall conform to grade DF-A of VV-F-800, except that grade DF-2 of VV-F-800 shall be used during the first 16 hours of endurance test (see 4.8.11).

3.4 Performance.

3.4.1 Heat output. When tested as specified in 4.8.2.3, the heater shall have heat output as specified in table II.

TABLE II. Heat output.

Capacity (see 1.2)	BTU's per hour +20%	
	Min. high setting	Min. low setting
A	18,000	8,000
B	27,000	14,000
C	54,000	30,000

3.4.2 Air discharge rate. During test specified in 4.8.3, heaters performing at either low or high heat shall discharge ventilating air at not less than rates specified in table III.

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TABLE III. Air discharge, ventilating.

Capacity (see 1.2)	Standard cubic feet per minute			
	Unrestricted		Air back pressure at 1.00 inch H_2O	
	Min.	Nom.	Min.	Nom.
A	70	80	33	39
B	115	130	80	94
C	200	225	150	174

3.4.3 Fuel flow. When tested as specified in 4.8.4, the fuel flow rate and pressure shall be as specified in table IV.

TABLE IV. Fuel flow.

Capacity (see 1.2)	High flow lbs./min.		Low flow lbs./min.		Pressure PSI (max.)	Pressure PSI (min.)
	Max.	Nom.	Max.	Nom.		
A	0.034	0.029	0.016	0.013	15	3
B	0.049	0.042	0.031	0.026	15	3
C	0.078	0.071	0.041	0.036	15	3

3.4.4 Overheat switch. Overheat switch shall break the circuit and cut off flow of fuel before the ventilating air is heated to a maximum temperature of 600 degrees Fahrenheit (°F) but not below 400°F. The switch shall be tested as specified in 4.8.6.

3.4.5 Low temperature operation. When tested as specified in 4.8.7, heaters shall have operational stability while conforming to 3.4.1.

3.4.6 Cycling. When tested as specified in 4.8.8, heaters shall start in less than 4 minutes and purge in less than 3 1/2 minutes.

3.4.7 Operating position. When tested as specified in 4.8.9, heaters shall start and operate without difficulty in any position of rotation about the longitudinal axis, and with the exhaust tube on or below the horizontal, while the axis is inclined at any angle from horizontal (0°) to vertical (90°), with blower intake end on or above the horizontal.

3.4.8 Vibration. When tested as specified in 4.8.10, heaters shall start and operate while vibrating. After testing, heaters shall have no evidence of fuel leakage and shall meet requirements of 3.4.

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3.4.9 Endurance. When tested as specified in 4.8.11, heaters shall operate without major failure and shall conform to 3.3.2, 3.4.1, 3.4.2, and 3.4.3. A major failure is defined as a breakdown of the heat exchanger, electric motor, combustion air blower wheel, ventilating air blower wheel, or fuel control valve. (Cleaning of fuel system screens and filters, replacement of the igniter and preheat register, and adjustment of the flame detector switch are permissible repairs within the intent of this requirement.) The life of the igniter shall average not less than 300 hours. The heater shall be capable of operation for 800 hours in all positions specified in 4.8.9.

3.4.10 Electromagnetic radiation. When tested as specified in 4.8.12, heaters shall meet the requirements of MIL-STD-461 and 462.

3.4.11 Corrosion resistance. When tested as specified in 4.8.13, heaters shall show no evidence of corrosion that would affect functional operation after subjection to a 96 hour salt solution spray.

3.4.12 Fuel leakage. When tested as specified in 4.8.14, with heaters in the "OFF" position, there shall be no visible leakage from heater fuel components during a five minute period.

3.4.13 Smoke density. When tested as specified in 4.8.15, heater exhaust gases shall contain no more than 10 percent smoke when measured with a Photovolt Corporation photoelectric smoke meter or approved equal.

3.5 Marking.

3.5.1 Identification marking. Unless otherwise specified, identification marking of heater equipment, assemblies, and service parts shall be in accordance with MIL-STD-130.

3.5.2 Name plate. The name plate, data, and instruction plates shall conform to the applicable provision of MIL-P-514.

3.6 Workmanship. Workmanship shall be of a quality which shall assure a product free of burrs, sharp edges, or other defects listed in table VI.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except

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as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. The manufacturer shall insure that test and inspection facilities of sufficient accuracy, quality, and quantity are established and maintained to permit performance of required inspections.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.4).
b. Quality conformance inspection (see 4.7).
c. Verification of qualification (see 4.6).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified.

4.3.1 Conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with test conditions specified in the "General Requirements" of MIL-STD-202.

4.4 Qualification inspection.

4.4.1 Qualification tests. The qualification sample shall consist of 2 heaters of each capacity specified. The samples shall be representative of the units proposed to be furnished under a contract. Qualification tests shall be conducted by the Government at a place designated by the Government. Heaters shall be examined for defects specified in table VI and tested as specified in table V. All tests with exception of 4.8.12, 4.8.13, and 4.8.16 shall be performed at a temperature of minus 65° $\pm$ 5°F.

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TABLE V. Order of qualification tests.

Sample 1		Sample 2	
4.8.2	Heat output	4.8.10	Vibration
4.8.3	Air discharge	4.8.6	Overheat switch
4.8.7	Low temperature	4.8.11	Endurance (including 4.8.7.1)
4.8.8	Cycling	4.8.2	Heat output
4.8.2	Heat output	4.8.3	Air discharge
4.8.3	Air discharge	4.8.14	Fuel leakage
4.8.5	Electrical	4.8.15	Smoke density
4.8.4	Fuel flow	4.8.13	Corrosion
4.8.9	Operating position	4.8.1	Operational
4.8.12	Radio interference suppression		

4.4.2 Failure. Failure of a qualification sample to pass any tests specified herein shall be cause for the Government to refuse to qualify the product, or to conduct additional testing until faults revealed by tests have been corrected.

4.5 Quality conformance inspection.

4.5.1 Inspection lot. A lot shall consist of all heaters of one type from an identifiable production period, submitted for acceptance at one time.

4.5.2 Sampling for examination. Samples for examination shall be selected in accordance with inspection level II of MIL-STD-105.

4.5.3 Sampling for acceptance testing. Samples for acceptance testing shall be selected in accordance with inspection level S-3 of MIL-STD-105.

4.6 Verification of qualification.

4.6.1 Examination. Each sample heater selected in accordance with 4.5.2 shall be examined to verify compliance with this specification. Examination shall include characteristics specified in table VI, for conformance to specified acceptable quality levels (AQL's) on the basis of percent defective.

4.6.1.1 Unclassified characteristics. All defects having no bearing on function, safety, interchangeability, or life, but which are considered departures from good workmanship, will be noted in writing. Workmanship deficiencies falling within this category and recurring in

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five consecutive lots of 10, within a 30 day period, will be added to the minor classification of characteristics of the specification with no increase in AQL's.

4.7 Quality conformance inspection.

4.7.1 Acceptance test. All heaters shall be subjected to test specified in 4.8.1.

4.7.1.1 Test failure. If a heater fails to pass the acceptance test specified herein, the Government inspector shall stop acceptance of subsequent lots until satisfactory evidence has been provided by the contractor to the Government that corrective action has been taken.

TABLE VI. Classification of defects.

Categories	Defects	Methods of inspection
Critical:	None defined	
Major:	AQL 1.0 percent	
101 Dimensions	Dimensional nonconformance	Visual and functional
102 Voltage	Nonconformance	Functional
103 Components	Parts damaged affecting their serviceability	Visual and functional
104 Leakage	Fuel or exhaust	Visual
105 Wiring	Defective	Functional
106 Motor	Malfunction	Functional
107 Controls	Missing	Visual and functional
108 Assembly	Incomplete or incorrect	Visual and functional
Minor:	AQL 2.5 percent	
201 Identification	Illegible or incorrect	Visual
202 Dimensions	Dimensional nonconformance not affecting interchangeability or capacity	Visual
203 Mounting brackets	Missing or loose	Visual
204 Bolts, nuts and screws	Missing	Visual

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TABLE VI. Classification of defects. - Continued

Categories	Defects	Methods of inspection
205 Finish or pan painted surfaces	Paint peeling, blistered, or sealing wrong color	Visual
206 Assembly	Improperly assembled	Visual and functional
207 Workmanship	Poor quality	Visual

4.7.2 Control test. For conductance of control tests, the Government shall select samples at the rate of 2 from each 200 units of the type produced, except that not more than 4 nor less than 2 units shall be selected in any 30 day period. The control test samples shall be identified as to production period, examined in accordance with MIL-STD-105 for defects specified in table VI, and subjected to tests specified in 4.8.6, 4.8.9, and 4.8.14.

4.7.2.1 Test failure. If the control test samples fail to pass any specified test, the Government inspector shall stop acceptance of subsequent lots until evidence has been provided to the Government by the contractor that faults revealed by the tests have been corrected.

4.8 Methods of inspection.

4.8.1 Operational test. Each heater shall be operated sufficiently to assure there is no malfunction in the operation of blowers and blower motor.

4.8.2 Heat output test.

4.8.2.1 Heat output equipment. The heat test equipment shall consist of a plenum chamber, outlet duct, thermocouples, and a nozzle set up to measure the air temperature rise and air flow through heater, thereby determining heat output.

4.8.2.2 Temperature. Thermocouples shall be installed at heater air inlet, and inlet air temperature shall be controlled to minus 65° $\pm$ 5°F. Heater air outlet temperature shall be determined by a series of thermocouples located in plenum chamber and at the nozzle. Air temperature rise through heater shall be the temperature difference of leaving and entering air.

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4.8.2.3 Heat output. Output of the heater shall be determined by the following formula:

$H = VdCp\ (T_2 - T_1)$

H = Heat output in British thermal units (Btu) per hour (hr.).

V = Air flow in cubic feet (cu. ft.) per hr.

d = Density of air.

Cp = Specific heat of air, 0.24 Btu per pound (lb.) per °F.

T_1 = Inlet air temperature in °F.

T_2 = Outlet air temperature in °F.

Testing of sample 1 shall be conducted at 24 ± .5 volts. Record measurements required to calculate heat output using the above formula to determine conformance to 3.4.1. Testing of sample 2 shall be conducted during the endurance test with voltage being used at the time, (see 4.8.11) and with back pressures up to and including maximums as follows:

Capacity (see 1.2)	Inches of H_2O Ventilating	Inches of H_2O Exhaust
A	0.50	0.50
B	0.75	0.75
C	1.00	0.75

4.8.3 Air discharge test. With heater in operation as stated in 4.8.2, air flow through heater shall be obtained by measuring the pressure drop across a bureau of standards type nozzle having a coefficient of discharge of 0.986. Testing of sample 1 shall be conducted at 24 ± .5 volts. Testing of sample 2 shall be conducted during the endurance test, and shall be conducted with voltage being used at the time (see 4.8.11). Calculate the air discharge using the following formula to determine conformance to 3.4.3:

$$Q = 1082\ A\sqrt{\frac{P}{d}}$$

Q = Air discharge in cu. ft. per minute (min.).
1082 = Constant (see 6.4.1.3)
A = Area of the nozzle in square feet (sq. ft.).
P = Static pressure in inches of water
d = Density of air in lbs. per cu. ft.

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4.8.4 Fuel flow test. A solenoid controlled 3-way valve shall be connected before the fuel pump with one valve inlet connected to a burette and the other inlet to the fuel supply. The solenoid valves shall be connected electrically to a timing device which will open the burette side of valve for 2 minutes. Fuel flow and pressure shall be checked for each fuel type specified in 3.3.3 at each respective temperature utilization point to determine conformance to 3.4.3. On sample 2, the test shall be conducted during endurance test, and shall be conducted with voltage being used at that time (see 4.8.11).

4.8.5 Electrical test. Heater shall be tested with starting voltage of 24 volts. Using no orifice restriction on fresh air outlet, heater shall be started at low heat setting with heater in normal position and operated for 15 minutes with purges. Heater shall be allowed to cool for 35 minutes. Heater shall then be set at high heat and starting procedure repeated. A trial of 6 runs shall be made, 3 at low heat and 3 at high heat, to obtain current values to determine conformance to 3.3.2.

4.8.6 Overheat switch test. The test shall employ a venturi, thermocouple, and ducts. Heater shall be mounted to exhaust ventilating air through a duct approximately 5 feet in length. A venturi and thermocouple shall be placed at duct outlet. To determine conformance to 3.4.4, heater shall be run on high heat, and ventilating air flow restricted by holding cardboard or sheet metal over duct or venturi outlet. When shutting off ventilating air supply, care must be taken to insure combustion air supply.

4.8.7 Low temperature test. To determine conformance to 3.4.5, heater shall be run for 100 hours on alternate high and low heat settings with a total of 70 hours on high heat. The test shall be conducted at 24 $\pm$ .5 volts.

4.8.7.1 Extreme low temperature test. Using the qualification samples (see 4.4.1), the heaters, after being exposed to temperatures of minus 65 $\pm$ 5°F, should start consistently 50 out of 50 attempts (100% compliance). This test should be run concurrently with endurance tests (see 4.8.11) allowing sufficient time between shutdown for heaters to reattain specified initial start test conditions.

4.8.8 Cycling test. Heater shall be cycled in normal operating position at low heat setting for 50 cycles. Each cycle shall consist of starting at low heat, running for 15 minutes, followed by purging and cooling the heater for 35 minutes. The starting time shall be measured from the time the heater is turned on until the igniter circuit is

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deenergized. Purge time shall be measured from time the heater switch is turned off until the blowers shut off. The average starting and purge time shall be recorded to determine conformance to 3.4.6. The test shall be conducted at 24 ± .5 volts.

4.8.9 Operating position test. To determine conformance to 3.4.7, the heater shall be started and run for 10 hours in each position specified herein. In all positions, the blower intake end shall be on or above the horizontal plane. The test shall be conducted in the following positions at 24 ± .5 volts:

	Longitudinal axis	Rotated position
(a)	Horizontal	Normal operating
(b)	Horizontal	90° turn clockwise from (a)
(c)	Horizontal	90° turn counter clockwise from (a)
(d)	Tilted 45° with air inlet end up	Exhaust outlet down
(e)	Tilted 45° with air inlet end up	90° turn clockwise from (d)
(f)	Tilted 45° with air inlet end up	90° turn counter clockwise from (d)
(g)	Vertical with air inlet end up	

4.8.10 Vibration test. Heater shall be vibrated at a double amplitude of 0.125 inch for 8 hours in the direction of each of the three major axes of heater at 20 cycles per second. After 1 hour of vibrating, the heater shall be started, run for 1 hour, then shut off (but vibrating continued); this procedure shall be followed for 8 hours in each axis. After vibrating on each respective axis, heater shall be checked for fuel leaks to determine conformance with 3.4.8.

4.8.11 Endurance test. Heater shall be mounted in normal horizontal position. Heater shall then be started and run for 400 hours, operating alternately for 24 hours at high and low settings, while cycling 20 minutes on and 10 minutes off. Fuel usage shall be as specified in 3.3.3. Outlet air temperature shall be limited to 350°F, maximum. Heater shall then be mounted in the vertical position with air inlet end up. Heater shall be started and run for an additional 400 hours to accumulate a total of 800 hours. The same conditions shall prevail in the vertical position as used in the horizontal position. Use "ON" time only for totaling the 800 hours. Heater shall be checked at 20 ± .5 volts, 24 ± .5 volts, and 28 ± .5 volts to determine conformance to 3.4.9. The endurance test shall be conducted at the following start voltages in each of the two mounting positions:

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100 hours at 20 ± .5 volts
100 hours at 24 ± .5 volts
200 hours at 28 ± .5 volts

4.8.12 Electromagnetic interference test. Heater shall be tested in accordance with MIL-STD-462 to verify conformance with 3.4.10.

4.8.13 Corrosion resistance test. Heater shall be subjected to salt solution spray test method 101A of MIL-STD-202 to determine conformance to 3.4.11.

4.8.14 Fuel leakage test. Apply 15 pounds per square inch (psi) fuel pressure at inlet to determine conformance to 3.4.12.

4.8.15 Smoke density test. Measure smoke density of heater to determine conformance to 3.4.13. Before test, adjust gauge on smoke meter to read correctly for clear air flow through smoke meter. Operate heater for a five minute minimum on high heat before taking readings of exhaust flow through smoke meter. Take 5 readings, at 1 minute intervals, to obtain average readings of smoke density. The test shall be conducted at 24 ± .5 volts.

4.8.16 Materials and process. The Government inspector shall at unscheduled intervals, inspect all materials and processes involved in preparation for delivery to determine conformance to requirements of section 5 and specifications referenced therein. Any deviation from specified requirements shall be cause for refusal of the Government inspector to accept subsequent lots until satisfactory evidence has been provided by contractor that corrective action has been taken.

4.8.17 Lot by lot inspection. Lot by lot inspection of packing and marking shall be performed by contractor in accordance with the following procedures.

4.8.17.1 Inspection lot information. A lot shall consist of all packs packed in accordance with a single level of protection (level A, B, or C) from an identifiable production period, submitted at one time for inspection.

4.8.17.2 Sampling inspection.

4.8.17.2.1 Visual dimensional and primary inspection. Sampling shall be in accordance with MIL-STD-105. The classification of defects and acceptable quality levels (AQL's) shall be as listed below:

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TABLE VII. Classification of defects.

Categories	Defects	Method of inspection
Major:	AQL 1.0% defective	
101	Container not as specified	Visual
102	Improper closure of container	Visual
103	Marking omitted, incorrect, or illegible	Visual
104	Preservation, incorrect	Visual
105	Weight excessive	Visual
Minor:	AQL 2.5% defective	
201	Closure improper	Visual
202	Strapping, improper	Visual
203	Container, defects	Visual
204	Coating, incorrect	Visual

4.8.17.3 First pack. The first pack packed with level A requirement shall be submitted to the Government for examination and tests specified herein.

4.8.18 Packaging and pack tests.

4.8.18.1 Packaging tests. To determine conformance to level A, first pack packaging (see 4.8.17.3) the package shall be subjected to tests specified for method II B of MIL-P-116.

4.8.18.2 Pack test. To determine conformance to level A, the first pack (see 4.8.17.3) shall be subjected to tests of MIL-STD-1186.

4.8.19 Failure. Failure of the first pack or packaging to pass any specified test shall be cause for the Government to refuse acceptance of the first pack until proved to the satisfaction of the Government that faults revealed by tests have been corrected.

5. PACKAGING

5.1 Packaging, packing, and marking. Cleaning, packaging, packing, and marking shall be in accordance with the applicable packaging data sheet furnished at the time of invitation for bids.

6. NOTES

6.1 Intended use. Heaters are intended for use on military vehicles or equipment which may be required to operate in an ambient temperature to minus 65°F, from which a supply of fuel and electrical

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power is available; and when heated air is required, without the vehicle engine being operated.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) The applicable capacity (see 1.2), and applicable specification sheet (see 1.2.1).
(c) If controls are to be furnished (see 3.3.2.2).
(d) Selection of method and applicable level of preservation, and packaging, and level of packing of the referenced specifications (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List, whether or not such products have actually been so listed by that date. The attention of contractors is called to this requirement, and manufacturers are urged to have the products they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the QPL is the Commanding General, US Army Tank-Automotive Command, ATTN: DRSTA-GSS, Warren, MI 48090, and information pertaining to qualification may be obtained from that activity.

6.4 Definitions.

6.4.1 Derivation of formula. The formula for V, velocity of air flow, utilized in 4.8.2.3 was derived as follows.

6.4.1.1 Head. The head in feet of air is obtained as follows:

$$h = \frac{62.31}{12d} \times P = \frac{5.19P}{d}$$

When:
h = head in feet of air.
d = density of air in lbs. per cu. ft.
P = pressure or head in inches of water.
62.31 = density of water in lbs. per cu. ft. at 70°F.

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6.4.1.2 Velocity. Velocity of air flow in cu. ft. per hour is obtained as follows:

$$V = 60 \times 60\sqrt{2\,gh}$$

$$= 60 \times 60\sqrt{\frac{2x\ 32.2\ x\ 5.19P}{d}}$$

$$= 65815.6\sqrt{\frac{P}{d}}$$

6.4.1.3 Air discharge. Air discharge formula, in cu. ft. per min., stated in 4.8.3 is derived as follows:

$$Q = C\ A\ v$$

$$= 0.986 \times A \times \frac{65815.6}{60}\sqrt{\frac{P}{d}}$$

$$= 1082\ A\sqrt{\frac{P}{d}}\quad \text{When}$$

C = Coefficient of discharge on nozzle (see 4.8.3).

A = Cross section area of nozzle in sq. ft.

v = Velocity of air flow in cu. ft. per min.

6.5 Recycled materials. The use of recycled materials, which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item, shall be encouraged (see 3.2).

6.6 Changes from previous issue. Asterisks are not used in this revision, to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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